RANDOMIZED CONTROLLED TRIALS (RCTs): STUDY DESIGN

STATISTICS 140

APRIL 1, 2026

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RCT gold standard evidence

Randomization is key

- Minimizes differences in prognostic factors associated with the outcome of interest (confounders)

Prospective protocol minimizes sources of variation between arms

- Strict inclusion & exclusion criteria
- Standardized treatment plans within each arm
- Uniform follow-up schedule
- Common panel of required tests and procedures e.g scans, labs
- Formal definition of endpoints

Bias in estimating treatment effect may still arise due to

- Missing data, treatment compliance, endpoint misspecification

PARTIQoL

Prostate Advanced Radiation Technologies Investigating Quality of Life

Stratify

- *Study site*
- *Age (<65 vs.* ≥ 65y)
- Intent to use rectal spacer (No vs. Yes)
- Moderate hypofrac (No vs. Yes)

Randomize 1:1

450 men

Proton Beam Therapy (PBT)

versus

Intensity Modulated Radiotherapy (IMRT)

Follow-up

Every 3 months for the first year, every 6 months for the second year, yearly after the second year to 5 years post-RT

Primary Endpoint

- Bowel Function at 24 mo. (EPIC)

Secondary Outcomes

- Urinary and erectile function
- HRQOL and Utilities
- Perceptions of care
- Adverse events
- Efficacy endpoints
- Direct and Indirect costs

	Intensity modulated radiation therapy (IMRT)	Proton Beam Therapy (PBT)
Radiation Modality	Photons (X-rays)	Protons (positively charged particles)
How It Works	X-ray beams from many angles, adjusting intensity to match the tumor shape	Proton beam that stops at the tumor, delivering radiation directly to the cancer and not beyond
Availability	Widely available in most cancer centers	Less common, only available in specialized proton centers
Cost	Covered by most insurance, generally less expensive	More expensive, may not be fully covered by all insurers

Comparative Dosimetry of IMRT versus PBT

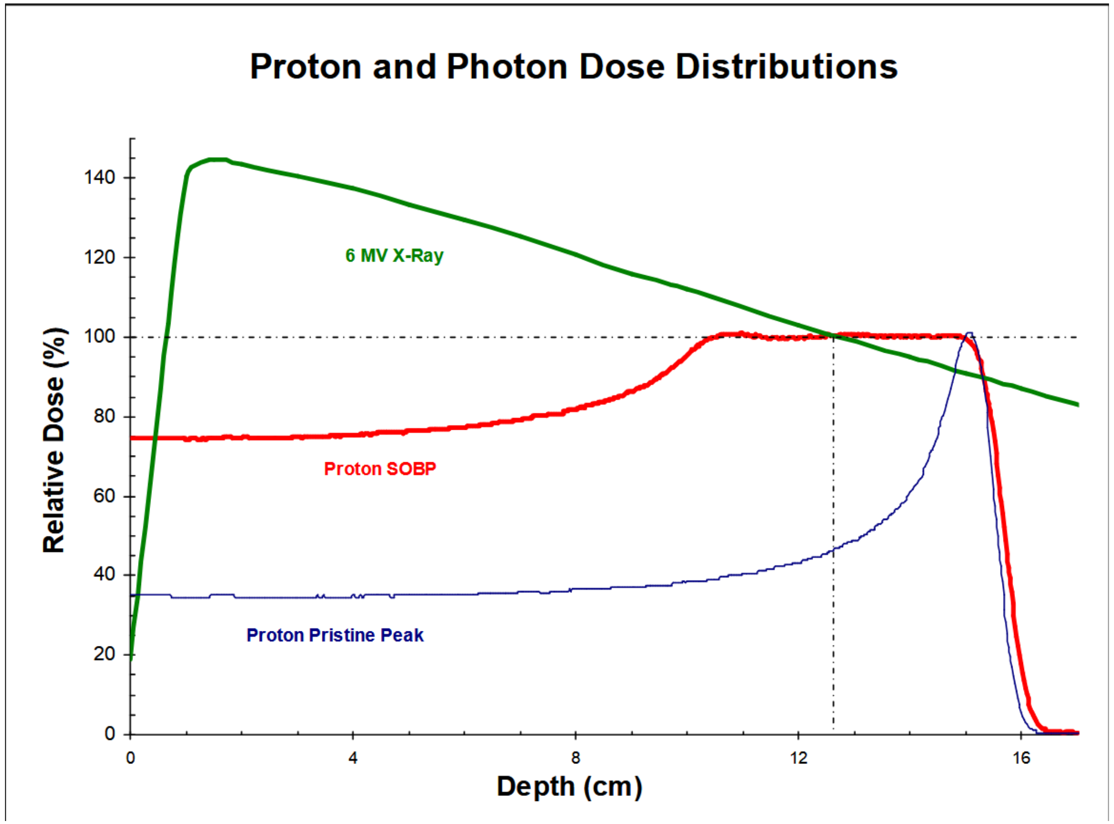
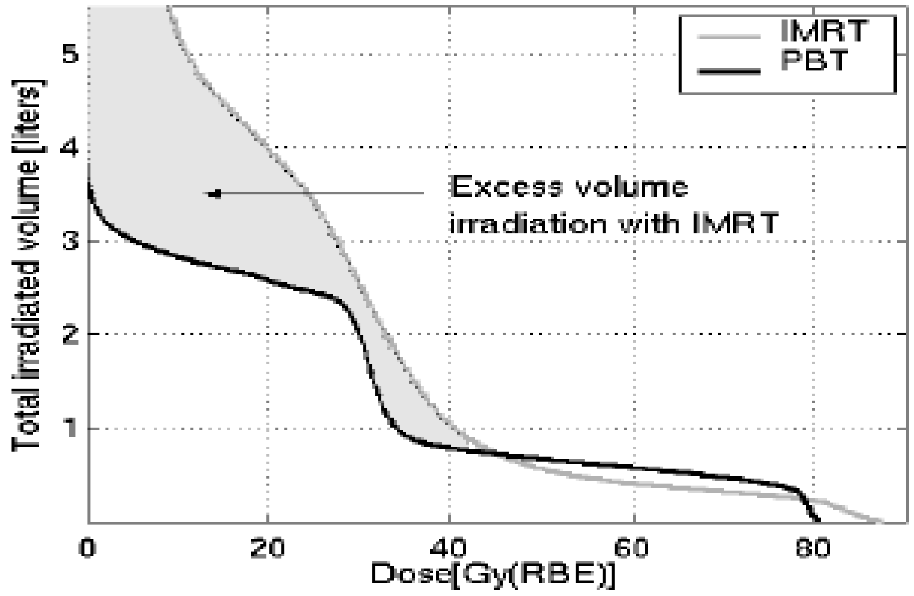


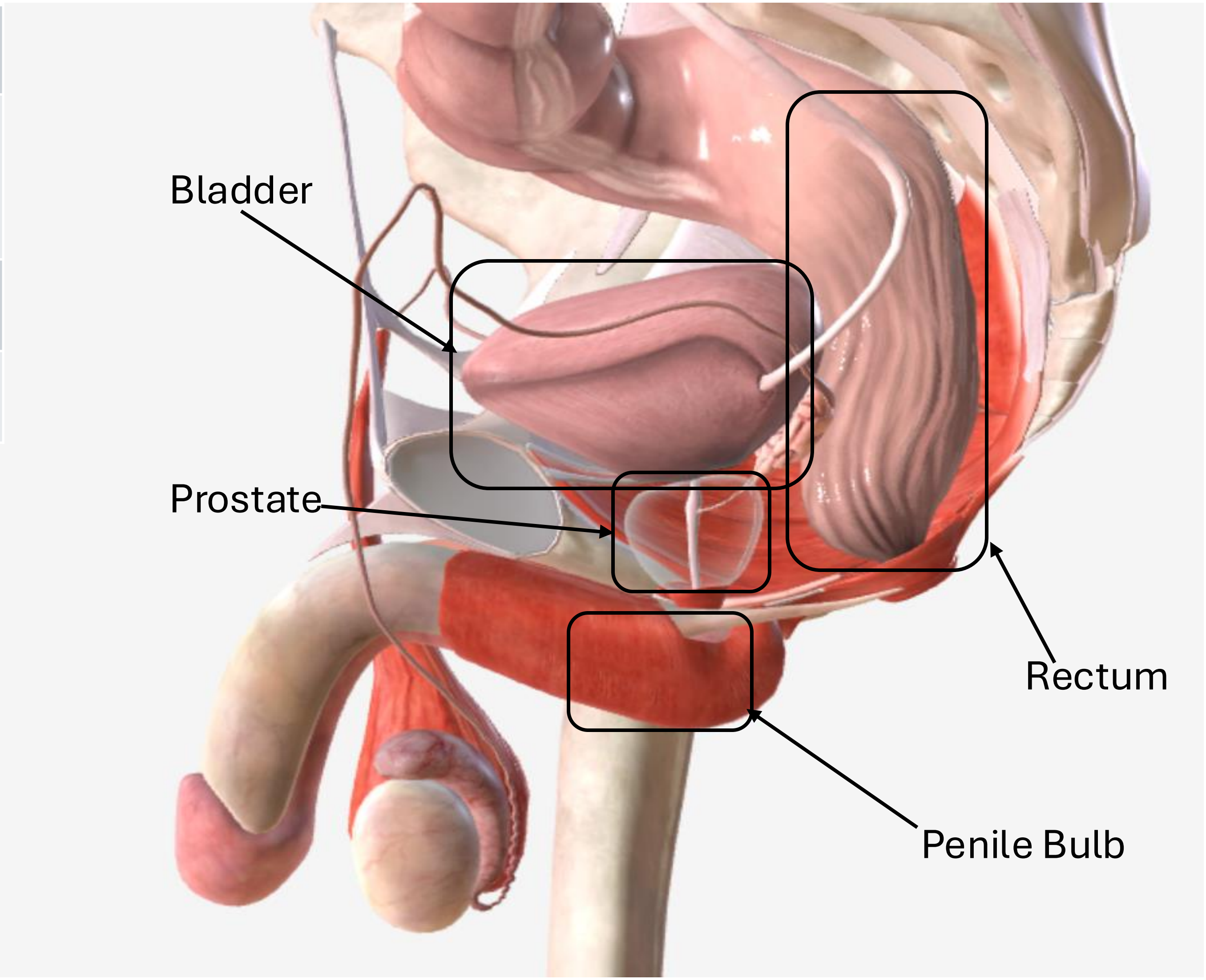
Figure 2. Difference in the Irradiated Volume Between IMRT and PBT for a Typical Patient



KEY ENDPOINTS

Organ at risk from radiation	Acute & Late Effects
Rectum	Bowel
Bladder	Incontinence Urinary Irritation
Penile bulb	Sexual
Prostate	Hormonal

- *Patient Reported Outcome (PRO): Expanded Prostate Cancer Index Composite (EPIC)*
- *Clinical Effects: Common Toxicity Criteria Adverse Events (CTCAE version 4.0)*



Clinical Hypothesis:
Proton beam therapy (PBT) will improve radiation-related effects compared to intensity modulated radiation therapy (IMRT) for a given radiation dose

- **Primary endpoint**: Bowel function (EPIC) at 24 months post-radiation
- Secondary PROs (EPIC): Urinary (incontinence, irritation), sexual, hormonal
- Adverse events: CTCAE version 4.0
- Efficacy endpoints:

    Disease-specific survival: Biochemical recurrence-free, metastasis-free, cancer-specific survival

    Overall survival

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- **Adverse events**
- **Efficacy endpoints**
- **Direct and Indirect costs**

Simple Randomization

No restrictions

- Depends only on final sample size and randomization ratio
  e.g. 1:1, 2:1, 1:1:1:1, 2:2:1
- Possible imbalances of patient numbers assigned between arms, especially when total sample size is relatively small (N<100)
- Possible imbalances of prognostic distribution between arms, especially when total sample size is relatively small (N<100)

Permuted Blocks Randomization

1. Divides patients into blocks over time
2. Balances patient numbers assigned between arms per block

- Guarantees patient numbers are balanced between arms at the end of study
- Unequal (random) block sizes to circumvent predictability of future assignments per block

Block size = 4

AABB	ABAB	ABBA
BBAA	BABA	BAAB

Stratified Randomization

Simple or permuted blocks randomization within unique groups defined by stratification factors

- Reduces prognostic imbalances between arms by balancing patient numbers assigned between arms in each stratum
- Cumbersome if stratifying by too many prognostic factors

  e.g. 4 prognostic factors with 2 groups each => 16 strata
- Possible imbalances of patient numbers assigned between arms when #strata is large relative to total sample size, especially if many strata have relatively few patients

Alternative to Stratification

Minimization makes treatment assignment with the goal of balancing the prognostic distribution between arms.

- Reduces imbalances of prognostic factors between arms
- Not truly random as new treatment assignments depends on the prognostic distribution of patients already assigned.

PARTIQoL

Prostate Advanced Radiation Technologies Investigating Quality of Life

Randomize 1:1
450 men

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- Efficacy endpoints
- Direct and Indirect costs

Sample Size Calculation

Total of N=450 patients randomized 1:1 to PBT or IMRT (225 per arm) provides 90% power to detect an effect size of 0.4σ using a two-sample t-test at a two-sided significance level of 0.05.

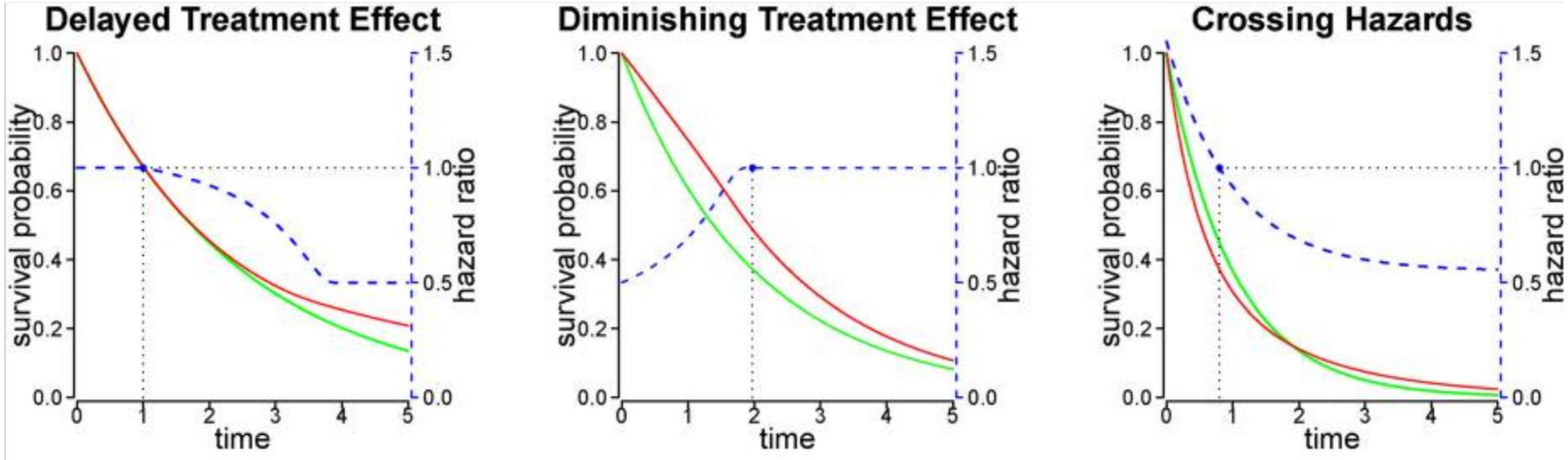
- σ represents standard deviation of the change in EPIC bowel score from baseline to 24 months.
- includes inflation factor to allow up to 37% of subjects lost to follow-up by 24 months.

Null and Alternative Hypotheses

- Null H_0: $\mu_{IMRT} – \mu_{PBT} = 0$
  (μ_{IMRT}, μ_{PBT} = mean of score change from baseline to 24 months)
- Alternative H_1: $\mu_{IMRT} – \mu_{PBT} = 0.4\sigma$
  (σ = sd of score change, assumed common)
- Two-sample t-test achieves 90% power for comparing the change in EPIC bowel score from baseline to 24 months between PBT and IMRT

Effect Size Specification

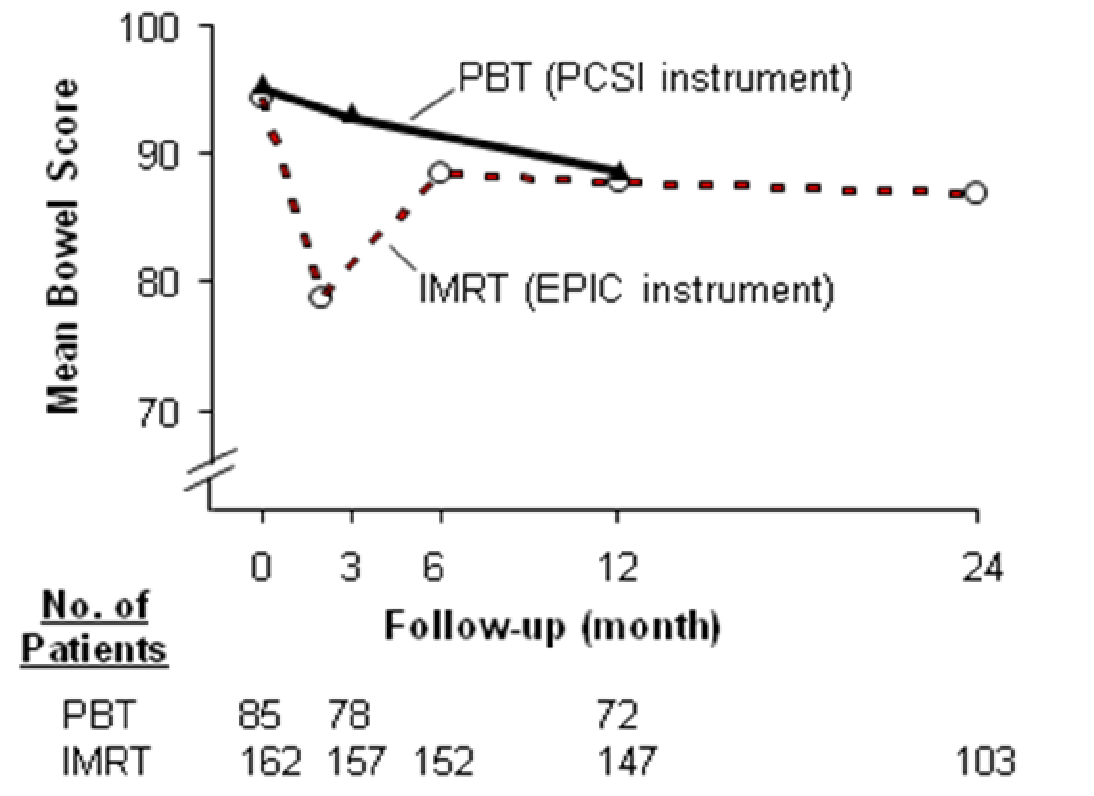
- Difference of means: $\mu_{IMRT} - \mu_{PBT}$
  t-test, Wilcoxon rank-sum
- Difference of proportions: $p_{IMRT} - p_{PBT}$
  Fisher's exact, c^2
- Difference of Survival functions: $S_{IMRT}(t) - S_{PBT}(t)$
  Log-rank, Cox model assuming proportional hazards (PH)
  Weighted log-rank, Restricted Mean Survival Time (RMST) under non-PH



Effect Size must be Clinically Meaningful and Realistic

- 0.5σ is the clinically important difference used in the validation cohort of the EPIC instrument.
- 0.5σ is clinically important difference suggested for assessing PROs.
- 0.4σ (~ 6±15 points) is the effect size for change from baseline to six months in pilot data.

Figure 3. Changes in Bowel Symptoms after PBT and IMRT



Power depends on Effective Sample Size

- Power = 1 – type 2 error β (false negative)
  = 1 – Prob of accepting null hypothesis of no difference
  when PBT and IMRT are truly different in bowel score change
  = Prob of rejecting null hypothesis in favor of accepting alternative
  when PBT and IMRT are truly different in bowel score change
- 37% rate of loss to follow-up at 24 months among 450 patients

=> Effective sample size of 280 patients (140 per arm)

90% power to detect an effect size of 0.4σ in bowel score change

Accrual & Effective Sample Size

- Accrual inflation:
  - Ineligible
  - Dropout before starting protocol treatment
  - Loss to follow-up
- Time to event or failure time endpoints e.g. survival
  - Power depends on #failure events (uncensored)
  - Higher accrual numbers with shorter follow-up duration
  - Lower accrual numbers with longer follow-up duration
- Effective sample size large enough to achieve 80-90% power in an RCT

Significance level α

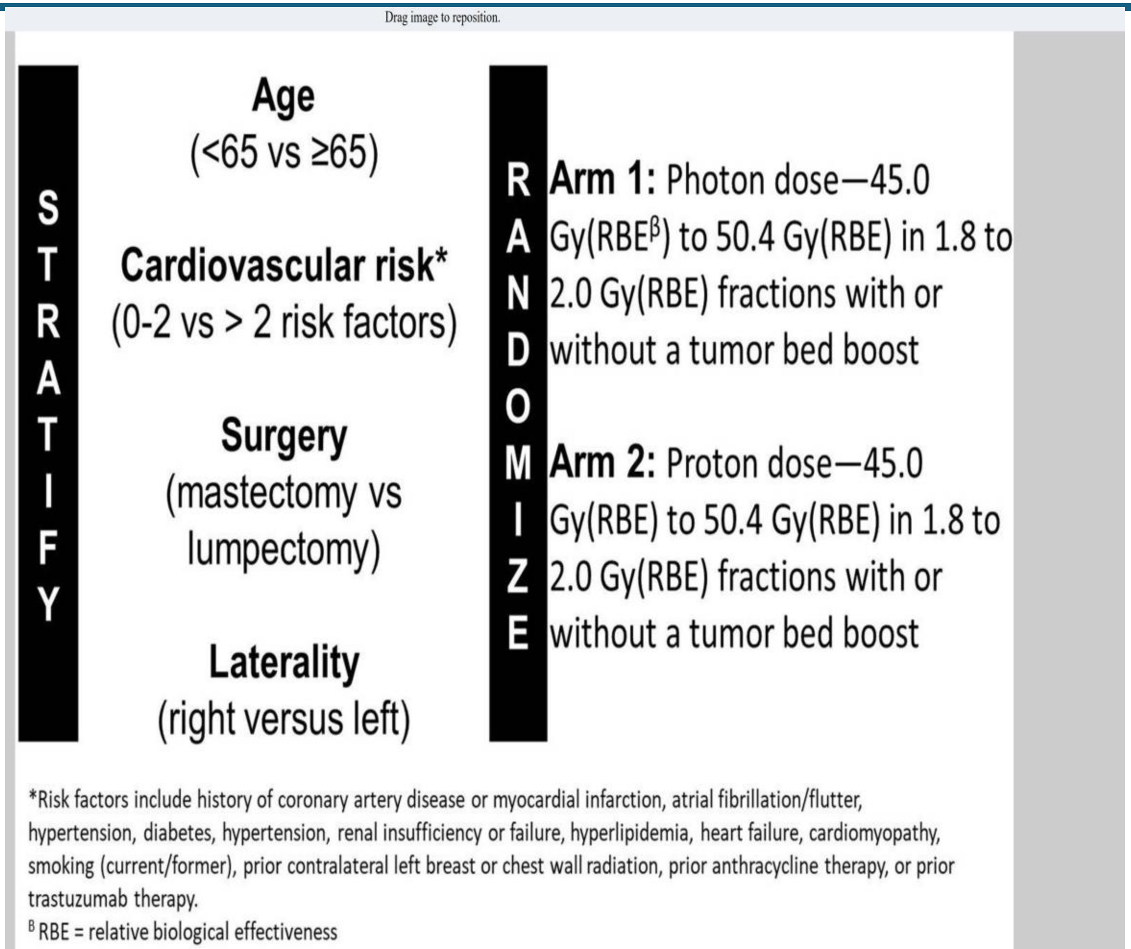
- α = type 1 error (false positive)
  = Prob of rejecting null hypothesis in favor of accepting alternative when PBT and IMRT are truly not different in bowel score change
- Two-sample t-test at a two-sided level of 0.05
- 0.05 two-sided level is the convention for RCTs
- α level could be smaller e.g. 0.01 or 0.005 for stronger statistical evidence

Two-sided or one-sided α?

- Two-sided: more conservative by protecting against unexpected results in either direction e.g. PBT better or worse than IMRT
  - Secondary, Exploratory, Unplanned Analyses
- One-sided:
  - Non-inferiority trials
  - Strong preliminary data
  - Superiority only of interest for regulatory purposes
  - Opposite direction is impossible e.g. placebo control
- One-sided:
  - Same sample size, more power than two-sided
  - Smaller sample size to obtain the same power as two-sided

Pragmatic Randomised Trial of Proton versus Photon Therapy for Patients with Non-Metastatic Breast Cancer: A Radiotherapy Comparative Effectiveness (RadComp) Consortium NCT02603341

- Total of N=1278 patients randomized 1:1 to photon or proton therapy (639 per arm) provides 80% power to detect 45% relative reduction of major cardiovascular events (MCEs) at 10 years to 3.5% for the proton arm using a log-rank test at a one-sided level of 0.05 (superiority hypothesis)
- 6.3% rate of MCEs for the photon arm based on Surveillance Epidemiological End Results (SEER) database
- Assume loss to follow-up rate of 13%
- 11/2015 to 07/2019: Total of 1720 patients



Co-Primary or Joint Primary Endpoints

- Trial is positive if null hypothesis were rejected in favor of the alternative for all co-primary endpoints
- Same level of type 1 error α for all hypothesis tests
- Power generally reduced due to increased risk of type 2 error β
- Co-primary endpoints

  PARTIQoL: score change in bowel PRO & urinary PRO

  Alzheimer's: cognition & functional outcomes

  Osteoporosis: fractures & bone density

  Irritable bowel syndrome (IBS): abdominal pain intensity & frequency of constipation or diarrhea

Multiple Primary Endpoints

- Trial is positive if null hypothesis were rejected in favor of the alternative for at least one primary endpoint
- Dual primary endpoints
  - PARTIQoL: score change in bowel PRO OR rate of bowel AEs
  - Organ transplantation: organ rejection OR survival rate
  - Osteoporosis: fracture rates at 12 months OR 24 months
- Multiple testing: overall control of study-wise type 1 error α

Hypothesis Testing for Multiple Primary Endpoints

Primary endpoints of equal importance

- Bonferroni adjustment: divide overall α by #tests

  Osteoporosis: α_1=0.025 for fracture rate at 12 months
  α_2=0.025 for fracture rate at 24 months

Primary endpoints with different priority

- Distribute α according to required level of evidence

  Oncology efficacy trials: α_1=0.005 for tumor response rate
  α_2=0.045 for survival endpoint

Gatekeeping Design for Multiple Primary Endpoints

Overall control of study-wise type 1 error a by comparing multiple endpoints sequentially in a pre-defined order and stopping all further testing once a non-significant result occurs.

- CLEAN-TAVI trial:

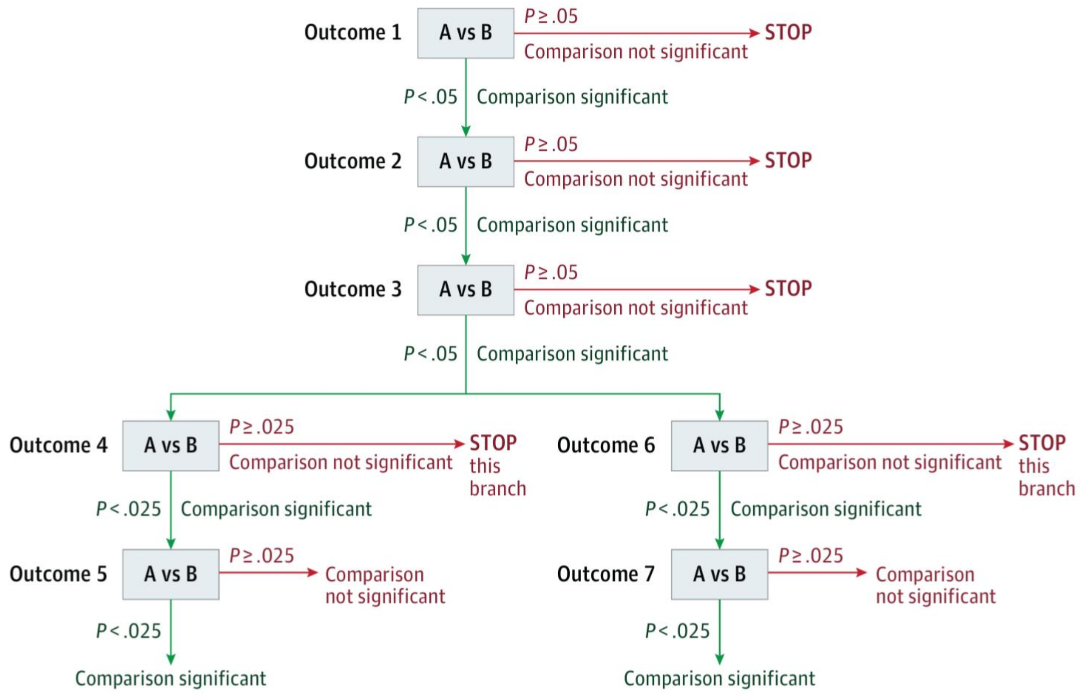
  Primary endpoint: #ischemic lesions 2 days post-procedure

  Secondary endpoints: 16 measures of number, volume, timing of cerebral lesions in various brain regions

Haussig S, Mangner N ,Dwyer MG, et al. Effect of a cerebral protection device on brain lesions following transcatheter aortic valve implantation in patients with severe aortic stenosis. JAMA. 2016; 316(6):592-601.

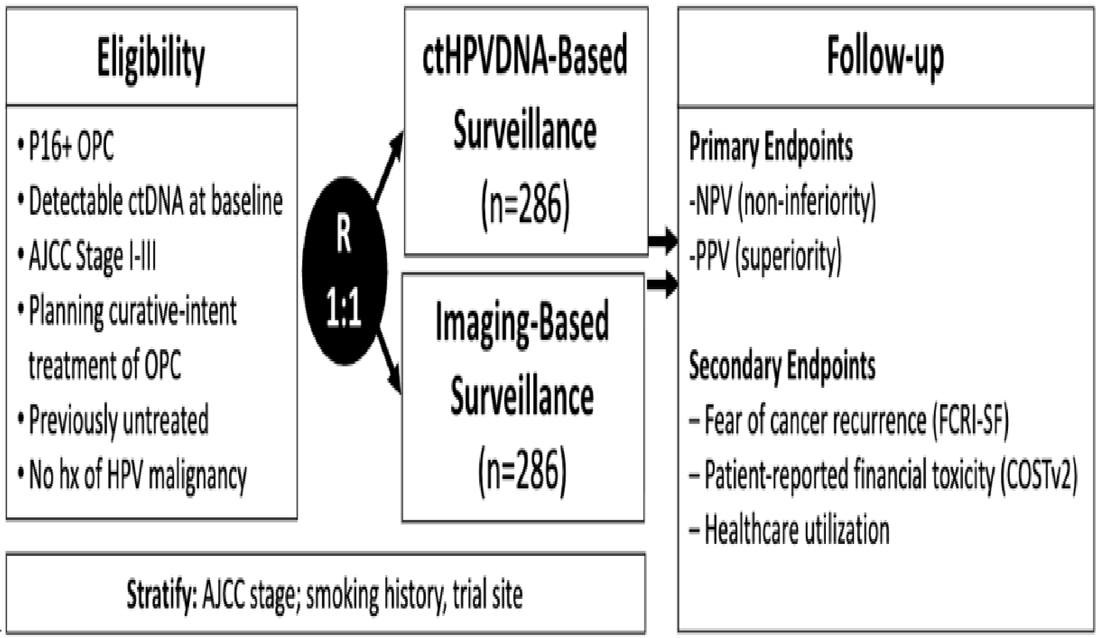
Statistical Significance Criteria for Hypothesis Testing in Gatekeeping Strategy

Yadav K, Lewis RJ. Gatekeeping Strategies for Avoiding False-Positive Results in Clinical Trials With Many Comparisons. In: Livingston EH, Lewis RJ. eds. *JAMA Guide to Statistics and Methods*. McGraw-Hill Education; 2019.



This Figure shows the criteria for statistical significance that would be used in a hypothetical gatekeeping strategy in which there are 3 levels each with a single end point, followed by 2 levels with 2 end points each. The 3 end points are each tested in order against a criterion of .05. All testing stops as soon as 1 result is nonsignificant. If all are significant then a pair of fourth-level end points are tested, and to preserve the required significance of .05 at that level across 2 end points, the criterion for statistical significance is adjusted with a Bonferroni correction value of .025 for each. If 1 or both of these end points is significant at .025, then the next end point in the branch is tested, against a criterion of .025. If 1 or both are nonsignificant, no further testing occurs. If any outcome tested along a given pathway is not statistically significant, no further outcomes along that branch are tested because they are assumed to be nonsignificant.

Figure 2



Interim Analysis in RCTs

- Efficacy: Stop trial early if strong evidence of effectiveness
- Futility: Stop trial early if very low probability of detecting a difference at trial completion
- Safety monitoring: Suspend trial temporarily or stop trial early if any arm is too toxic or risky
- Monitor sample size assumptions, potential study re-design:
  - Accrual rate
  - Randomization allocation
  - Distribution of key covariates: strat factors, confounders
  - Follow-up compliance
  - Event rate

Interim Analysis in RCTs

- Interim analysis must be pre-planned as part of study design to avoid inflation of type 1 error from repeated testing.
  - Group sequential design e.g. O'Brien-Fleming, Pocock
- Saves trial resources and time
- Protects trial patients from ineffective or unsafe treatments
- Expedite development of new therapies and dissemination to the general population

NEJM Evidence

Published June 27, 2023
NEJM Evid 2023; 2 (7)
DOI: 10.1056/EVIDe2300124

EDITORIAL

Potential Benefits and Risks of Interim Analyses in Clinical Studies

Tao Liu, Ph.D.[1]

Adaptive RCT Design

Use accumulating information during the trial to modify study design or trial conduct

- Adaptive randomization
    - Revise randomization ratio based on observed outcome to date
    - Assign more patients to more promising arm
- Pre-specify adaptive strategies with statistical criteria to ensure implementation is based on prospective and objective evidence
    - Requires timely update of outcome and relevant data
    - Continous data QA to ensure accurate information for decision making

Multiple Arms (>2) in RCT

- Multiple experimental arms vs. single control
- Single experimental arm vs. multiple controls e.g. active control and placebo control
- Platform design: experimental and control arms open or close throughout the trial duration based on efficacy decision rules
    - Discontinue ineffective treatments sooner
    - Minimize #patients treated with ineffective treatments
    - Add new promising treatments at any point
    - Assess more experimental treatments within a shorter period
    - Assign more patients to experimental arms than control arms

Cluster RCT Design

Randomize groups of subjects, not individuals

- More common in healthcare delivery, educational research
  - Hospitals, group practices, schools
- Administrative convenience, application ease, ethical considerations to deal with whole clusters
- Avoid contamination across individuals who receive different interventions within each cluster
- Power depends on #clusters and cluster size

Reporting of RCTs

- Intent-to-treat: primary analysis
  Analyze all patients per their assigned group regardless of their actual treatment
  Minimizes bias due to lack of treatment adherence
- Per-protocol: secondary analysis
  Analyze only patients who completed treatment per their assigned group
  Purer estimate of treatment effect?
  Potential bias due to lack of treatment adherence
- Treatment effect heterogeneity: different effects in subgroups
  Pre-specify subgroup analysis to avoid inflating type 1 error α

CONSORT (Consolidated Standards of Reporting Trials)

Hopewell S, Chan AW, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. **CONSORT 2025 statement: updated guideline for reporting randomized trials.**

- The Lancet, 2025; 405, 1633-1640
- *BMJ* 2025;389:e081123
- PLoS Med. 2025 Apr 14;22(4):e1
- *JAMA*. 2025;333(22):1998–2005
- *Nat Med* **31**, 1776–1783 (2025)

SUMMARY POINTS

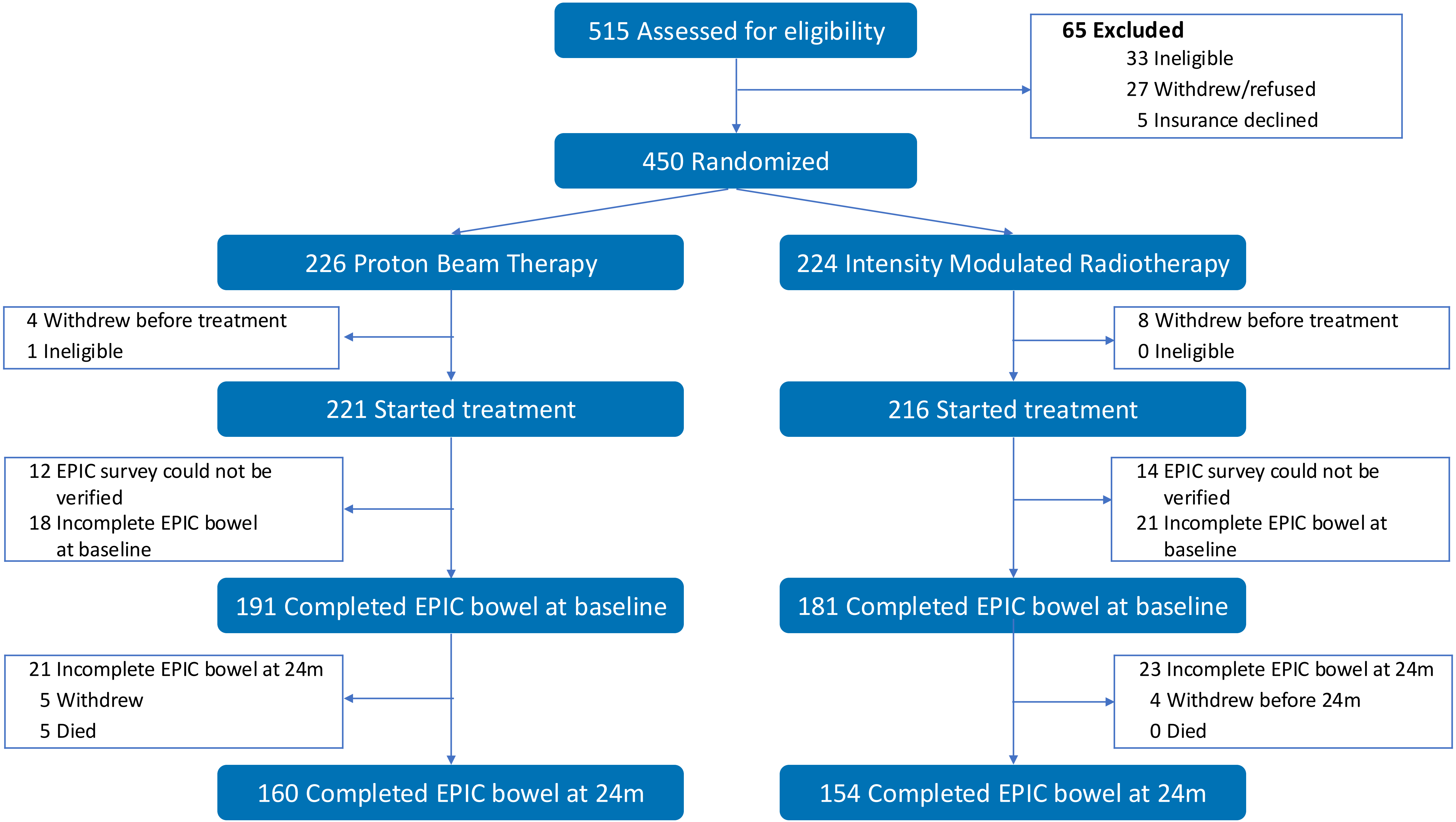
To interpret a randomised trial accurately, readers need complete and transparent information on its methods and findings

The CONSORT 2025 statement provides updated guidance for reporting the results of randomised trials, that reflects methodological advancements and feedback from end users

The CONSORT 2025 statement consists of a 30-item checklist of essential items, a diagram for documenting the flow of participants through the trial, and an expanded checklist that details the critical elements of each checklist item

Authors, editors, reviewers, and other potential users should use CONSORT 2025 when writing and evaluating manuscripts of randomised trials to ensure that trial reports are clear and transparent

the bmj | *BMJ* 2025;389:e081123 | doi: 10.1136/bmj-2024-081123


515 Assessed for eligibility
65 Excluded
33 Ineligible
27 Withdrew/refused
5 Insurance declined
450 Randomized
226 Proton Beam Therapy
224 Intensity Modulated Radiotherapy
4 Withdrew before treatment
1 Ineligible
8 Withdrew before treatment
0 Ineligible
221 Started treatment
216 Started treatment
12 EPIC survey could not be verified
18 Incomplete EPIC bowel at baseline
14 EPIC survey could not be verified
21 Incomplete EPIC bowel at baseline
191 Completed EPIC bowel at baseline
181 Completed EPIC bowel at baseline
21 Incomplete EPIC bowel at 24m
5 Withdrew
5 Died
23 Incomplete EPIC bowel at 24m
4 Withdrew before 24m
0 Died
160 Completed EPIC bowel at 24m
154 Completed EPIC bowel at 24m

Limitations of RCTs

Patient population

- Homogenous due to strict inclusion & exclusion criteria in RCT protocol
- Representative of the broader patient population?
- Generalizability of RCT results may be limited

Clinical settings

- Close monitoring of clinical status during RCT follow-up
- Extensive supportive care available in RCT institutions
- RCT environments rarely reflect real-world conditions

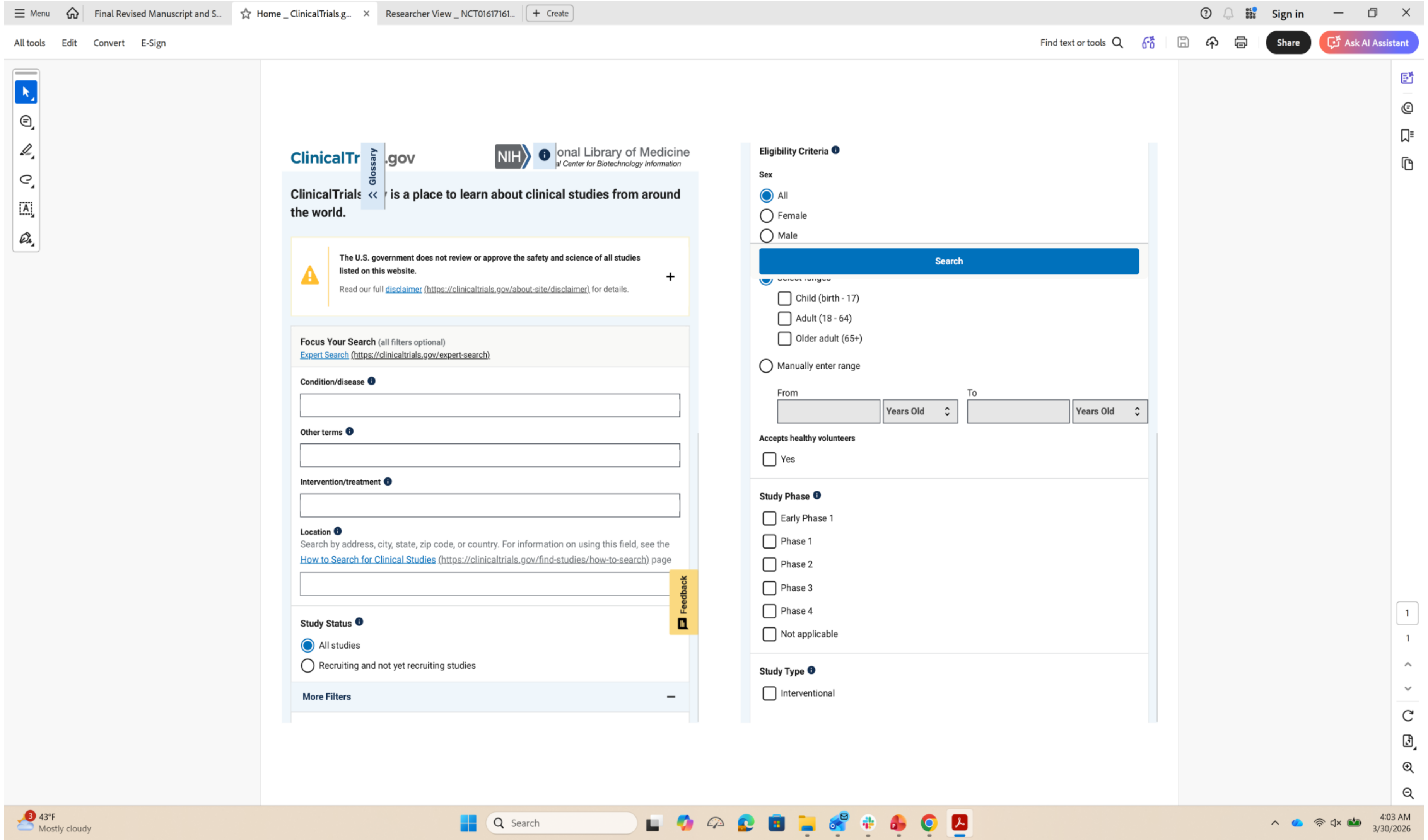
Barriers to RCTs

- Resource intensive: Cost, Time burdens
  - Trialists: Investigators, clinicians, research & technical staff
  - Sponsors: Funders, Institutions (IRB, DSMB), Regulators (FDA)
  - Patients, Families, Support networks
  - Healthcare Systems: Non-trial caregivers & administrators
- Ethical challenges in special populations
  - Children
  - Informed consent barriers
- Infeasible/Unfeasible
  - Very rare diseases
  - Referral patterns

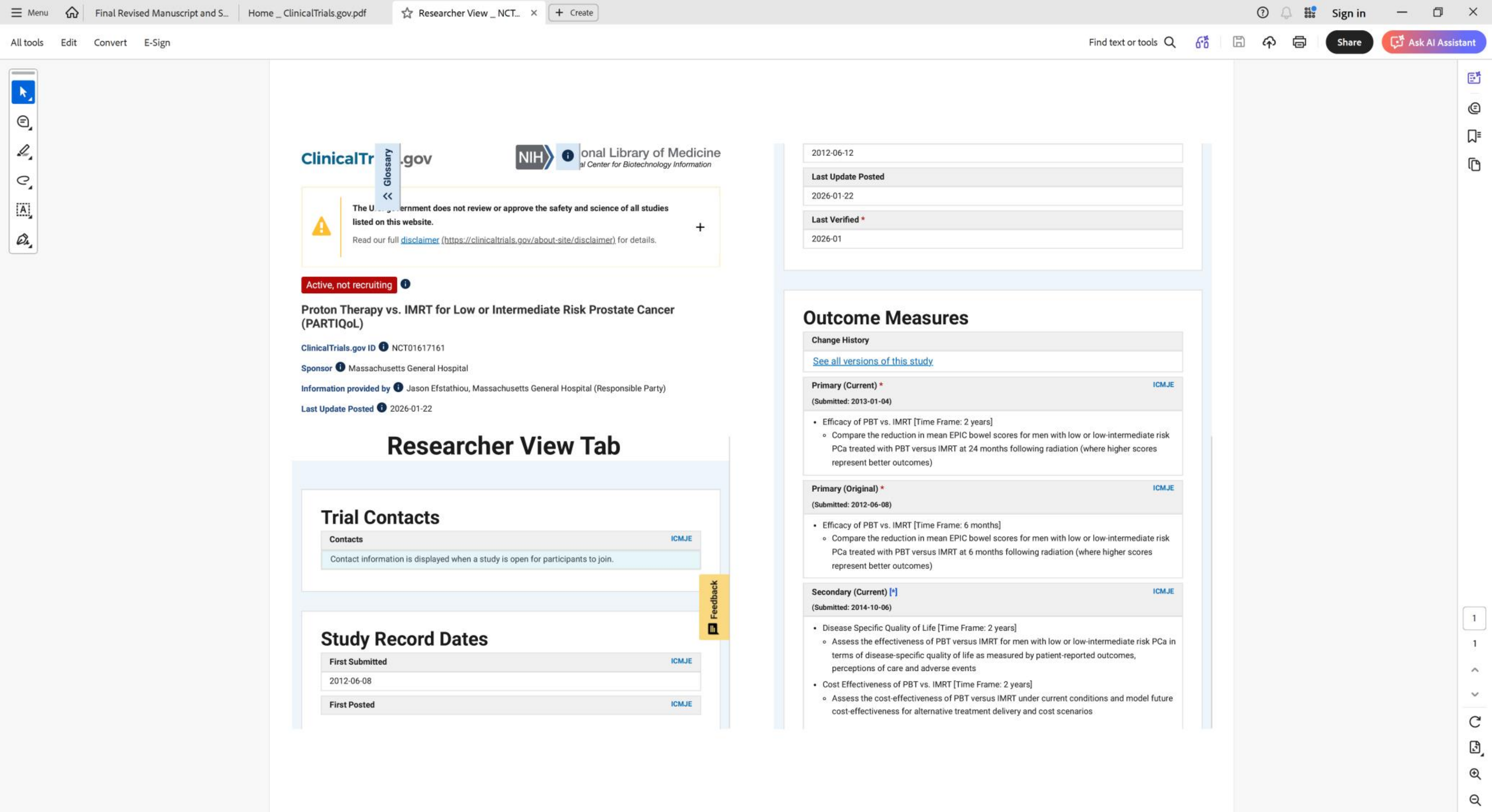
Major Milestones

- Mass General Brigham Mass General Cancer Center — First site activated Jun 2012
- 1st patient 1st visit Jul 2012
- Penn — Jan 2013
- Dec 2014 — Washington University in St.Louis
- CentraState / Northwestern Medicine Chicago Proton Center — Feb 2015
- Sep 2015 — University of Washington 1861
- Rutgers Cancer Institute of New Jersey, Rutgers Health — Mar 2016
- Apr 2016 — Provision Healthcare
- Amendment: Rectal Spacers Jun 2016
- Amendment: Hypofractionation Dec 2016
- Mayo Clinic — Jan 2017
- Feb 2017 — University of Maryland
- UFHealth Proton Therapy Institute — Dec 2017
- Dec 2018 — University Hospitals
- Amendment: Accrual to 450 Dec 2019
- Last patient 1st visit Nov 2021
- Primary Endpoint reached: 24-month follow-up Apr 2024
- Last patient last visit Feb 2034 (anticipated)

ClinicalTrials.gov



PARTIQoL (NCT01617161)



ClinicalTrials.gov

National Library of Medicine
National Center for Biotechnology Information

The U.S. government does not review or approve the safety and science of all studies listed on this website.

Read our full disclaimer (https://clinicaltrials.gov/about-site/disclaimer) for details.

Active, not recruiting

Proton Therapy vs. IMRT for Low or Intermediate Risk Prostate Cancer (PARTIQoL)

ClinicalTrials.gov ID NCT01617161

Sponsor Massachusetts General Hospital

Information provided by Jason Efstathiou, Massachusetts General Hospital (Responsible Party)

Last Update Posted 2026-01-22

Researcher View Tab

Trial Contacts

Contacts	ICMJE
Contact information is displayed when a study is open for participants to join.	

Study Record Dates

First Submitted	ICMJE
2012-06-08	
First Posted	ICMJE
2012-06-12	
Last Update Posted	
2026-01-22	
Last Verified *	
2026-01	

Outcome Measures

Change History

See all versions of this study

Primary (Current) * ICMJE
(Submitted: 2013-01-04)

- Efficacy of PBT vs. IMRT [Time Frame: 2 years]
  - Compare the reduction in mean EPIC bowel scores for men with low or low-intermediate risk PCa treated with PBT versus IMRT at 24 months following radiation (where higher scores represent better outcomes)

Primary (Original) * ICMJE
(Submitted: 2012-06-08)

- Efficacy of PBT vs. IMRT [Time Frame: 6 months]
  - Compare the reduction in mean EPIC bowel scores for men with low or low-intermediate risk PCa treated with PBT versus IMRT at 6 months following radiation (where higher scores represent better outcomes)

Secondary (Current) [*] ICMJE
(Submitted: 2014-10-06)

- Disease Specific Quality of Life [Time Frame: 2 years]
  - Assess the effectiveness of PBT versus IMRT for men with low or low-intermediate risk PCa in terms of disease-specific quality of life as measured by patient-reported outcomes, perceptions of care and adverse events
- Cost Effectiveness of PBT vs. IMRT [Time Frame: 2 years]
  - Assess the cost-effectiveness of PBT versus IMRT under current conditions and model future cost-effectiveness for alternative treatment delivery and cost scenarios

ASTRO's 66th Annual Meeting

Late-Breaking Abstracts

LBA 01

Prostate Advanced Radiation Technologies Investigating Quality of Life (PARTIQoL): Phase III Randomized Clinical Trial of Proton Therapy vs. IMRT for Localized Prostate Cancer

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ıwestern Medicine, Feinberg
of Radiation Oncology, Uni-
er Center, Seattle, WA,
nic, Rochester, MN, [9]Depart-
ryland School of Medicine,

IMRT and proton therapy offer equally high quality of life and tumor control for people with prostate cancer

Multi-center, phase III trial finds photon- and proton-based radiation therapies similarly safe and effective

WASHINGTON, September 30, 2024

People with low- and intermediate-risk prostate cancer treated with either of two types of contemporary radiation therapy — proton beam therapy or intensity modulated radiation therapy (IMRT) — achieved equally high rates of tumor control with no differences in patient-reported quality of life, according to a first-of-its-kind phase III clinical trial comparing the two technologies. Findings of the PARTIQoL trial will be presented today at the American Society for Radiation Oncology (ASTRO) Annual Meeting.

hypofractionation, 48% used a rectal spacer, and 49% of PBT patients were treated with pencil beam scanning. There was no difference between PBT or IMRT in mean change of health care software bowel score at 24 mo (p=0.836), with both arms showing only small, clinically non-meaningful decline from baseline (see Table). Similarly, there was no difference in bowel function at earlier timepoints (3, 6, 9, 12, 18 mo) or later timepoints (36, 48, 60 mo). No differences were observed in other domains (urinary, sexual, hormonal) at any timepoint. There was no difference in progression-free survival (PFS) (93.4% vs 93.7% at 60 mo, HR 1.16 [0.53, 2.57], p=0.706). There was no sustained difference in any QOL domain or PFS between arms in subgroups defined by stratification variables.

Conclusion: This prospective randomized clinical trial shows that patients treated with contemporary radiotherapy for localized prostate cancer achieve excellent QOL with highly effective tumor control, without measurable differences between PBT and IMRT. We continue to monitor participants for longer follow-up and secondary endpoints.

LBA 01 – Table 1

Health care software bowel score Mean (Std Dev)	PBT (N=221)	IMRT (N=216)	p-value
Baseline	93.7 (7.8)	93.5 (7.9)	
24 mo	91.8 (11.1)	91.9 (8.6)	